

Capstone Project - ResearchRAG - Research Paper Assistant.

1. Problem Statement

Researchers and students spend significant time searching, reading, and analysing research papers. Traditional keyword-based search may not understand the context and meaning of the content. Extracting information such as methodology, datasets, results, limitations, and research gaps from multiple papers manually is time-consuming.

ResearchRAG aims to provide an AI-powered assistant that allows users to upload research papers and ask questions in natural language. The system retrieves relevant content from the papers and generates context-aware answers using **Retrieval-Augmented Generation (RAG)**.

2. Problem Understanding

Researchers and students often need to read and analyse multiple research papers to understand existing work, methodologies, datasets, results, and research gaps. However, research papers can be lengthy and contain extensive technical information, making it difficult and time-consuming to locate specific details.

Traditional keyword-based search methods mainly look for matching words and may not understand the **context or meaning** of a user's question. For example, finding information about a paper's methodology or dataset may require manually searching through several pages.

Analysing multiple papers is even more challenging. Researchers may need to compare different approaches, identify common datasets, study experimental

results, and understand the limitations of existing work. Performing these tasks manually requires significant time and effort.

Large Language Models (LLMs) can provide natural-language answers, but they may sometimes generate information that is not supported by the original research papers. This is known as **hallucination**, which can be problematic in academic research.

ResearchRAG – Research Paper Assistant addresses these challenges using **Retrieval-Augmented Generation (RAG)**. Users can upload research papers in PDF format. The system extracts the text, divides it into smaller chunks, and converts the content into embeddings that are stored in a vector database.

When a user asks a question, the system retrieves the most relevant sections from the papers and provides them as context to an LLM. The LLM then generates an answer based on the retrieved information.

Thus, ResearchRAG helps researchers **quickly find relevant information, understand research papers, and reduce the time required for literature analysis.**

3. Problem Solution

ResearchRAG uses a **RAG-based approach** to retrieve relevant information from research papers and generate accurate responses.

Workflow:

1. Upload research papers in PDF format.
2. Extract and preprocess the text.
3. Divide the text into smaller chunks.
4. Generate embeddings for the chunks.
5. Store embeddings in a vector database.
6. Accept user questions in natural language.
7. Retrieve the most relevant paper sections.
8. Provide the retrieved content to an LLM.
9. Generate an answer with relevant source references.

4. Prerequisites

Software Requirements

- Python 3.10+
- VS Code / Jupyter Notebook
- Git and GitHub
- PDF processing libraries
- LangChain or LlamaIndex
- FAISS or ChromaDB
- LLM API or locally hosted LLM
- Embedding model
- Streamlit / Flask / FastAPI

Knowledge Prerequisites

- Basic Python programming
- Natural Language Processing (NLP)
- Machine Learning fundamentals
- Embeddings and vector databases
- Large Language Models (LLMs)
- Retrieval-Augmented Generation (RAG)
- Basic API and web development

5. Use Cases

- **Research Paper Q&A:** Ask questions and get answers from uploaded papers.
 - **Paper Summarisation:** Generate concise summaries of research papers.
 - **Research Gap Identification:** Identify limitations and potential research areas.
 - **Methodology Comparison:** Compare methods used across multiple papers.
 - **Dataset Analysis:** Find datasets, metrics, and experimental results.
 - **Literature Review:** Assist in analysing and organising multiple research papers.
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6. System Architecture

Research Papers (PDF)

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Text Extraction

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Preprocessing & Chunking

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Embedding Model

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Vector Database

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Semantic Search

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User Question

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Relevant Paper Chunks

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LLM

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Answer + Sources

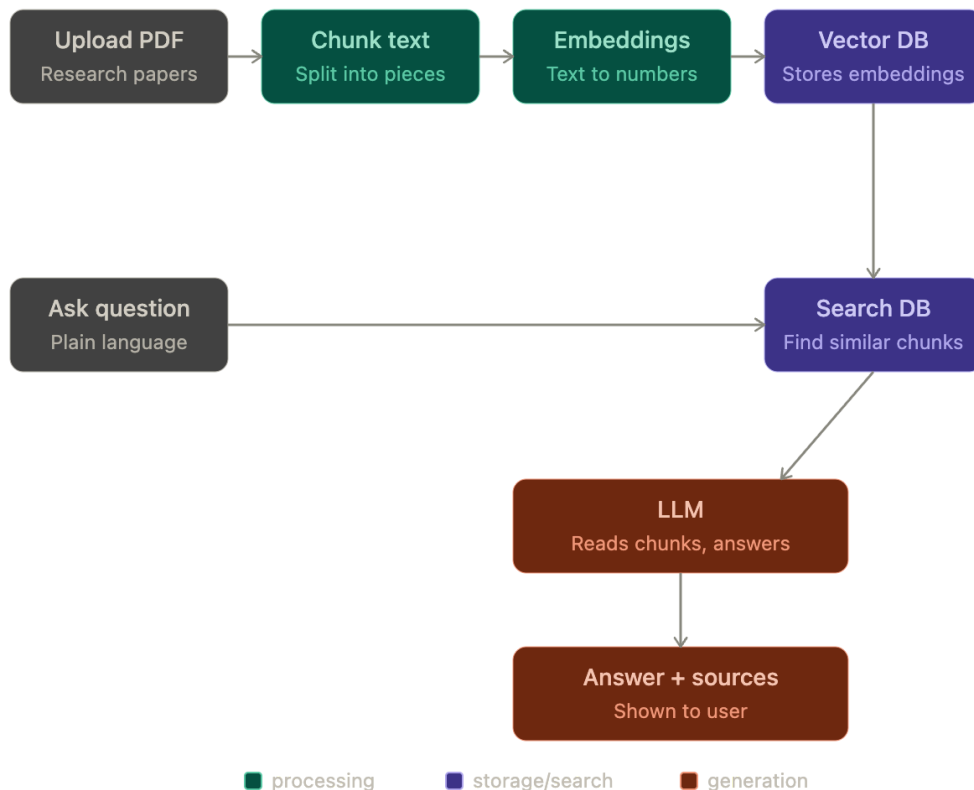
7. Limitations

- PDF extraction may be inaccurate for tables, images, and complex layouts.
- Retrieval may return irrelevant information for ambiguous queries.
- LLMs can still produce incorrect or hallucinated responses.
- Large collections of papers may require significant processing.
- API-based LLMs may involve additional costs.

8. Future Scope

- Support multimodal content such as tables, figures, and equations.
- Analyse and compare multiple research papers.
- Automatically identify research gaps and trends.
- Generate literature reviews and citation formats.

9. System Design



Project Summary

ResearchRAG – Research Paper Assistant is an AI-powered research assistant that uses **Retrieval-Augmented Generation (RAG)** to help researchers understand and analyse research papers. It extracts paper content, creates embeddings, stores them in a vector database, retrieves relevant information based on user queries, and uses an LLM to generate context-aware answers with supporting sources.